

Benjamin Pilgrim

ben@pilgrim.me.uk 

07343 061225 

Colchester, Essex 

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RISQ Research Senior DevOps Engineer

Feb 2024 - Present

Developed a custom CI/CD framework in Python for fast, reproducible local and GitHub Actions builds and deployments, instrumented with OpenTelemetry, underpinned by open-source libraries. [pyhcl2](#)  [pyagnostics](#) 

Delivered a custom AWS Well-Architected landing zone, leveraging account isolation boundaries for security and compliance controls using Terraform and AWS Organizations.

Designed IAM architecture for humans, workloads, pipelines, and cloud and co-located on-premises instances, enforcing least privilege through ABAC, OIDC, and permission boundaries.

Planned, implemented and administered RISQ Group's Information Security Management System, to comply with external standards within a regulated industry.

Architected RISQ's global hybrid network using Cloud WAN, IPAM, VPNs, and Route 53 for DNS, connecting cloud and co-located environments.

Defined reusable Terraform modules implementing consistent patterns for golden-path building blocks. These include short-lived scheduled and event-driven workloads managed through Step Functions and ECS, as well as stateful persistent low-latency trading systems run as systemd services, deployed via custom SSM associations.

Built a data platform for batch and streaming time-series market data, with metadata-driven integrity and retry handling.

Monitored trading infrastructure using CloudWatch and Grafana, with structured logs centralised to S3 and metrics dashboards providing operational visibility.

Designed and developed a customer facing gaming web application using React and Convex under tight timelines.

Saved £20,000+ in AWS spend through infrastructure and workload cost optimisation.

Designed and supervised ETL pipelines managing weekly transactions worth over £6,000,000 with human-in-the-loop approval.

MMT Digital Cloud Systems Engineer Vodafone Senior DevOps Engineer

Mar 2022 - Nov 2023

One of nine finalists for Young Digital Professional of the Year in the 2023 Computing Digital Technology Leaders Awards.

Drafted into a hand-selected specialist team to drive strategy and resolve key blockers on a global Internal Developer Platform.

Led development of control-plane interfaces for the platform, built on serverless technologies.

Led design, prototyping, and development of an Azure DevOps Pipelines framework for standardised multi-environment service delivery, deploying Terraform infrastructure and Helm charts to Kubernetes clusters.

Reduced time to first deployment for new services by 95%+, enabling teams to provision secure-by-design parameterised repository templates including pipeline definitions, Terraform infrastructure, Helm charts, and baseline application code.

Maintained globally adopted Terraform modules for AWS components, including API Gateway and static site hosting, driving efficiency through reuse.

Used Datadog to investigate and resolve production issues across the platform, spanning metrics, distributed traces, and logs.

Improved team code quality through adoption of automated testing, linting and type-checking, and mentored junior engineers through pair programming and code reviews.

Operated in a multicloud environment including GCP, resolving production client onboarding issues in Apigee API Gateway.

HTK Software Engineer

Jul 2019 - Mar 2022

Contributed as a backend engineer to distributed systems including a real-time transaction and promotion rules engine and a multi-tenant loyalty and marketing automation platform.

Developed third-party integrations and ETL pipelines for customer transaction data, including POS systems.

Delivered a multi-account AWS landing zone proof-of-concept in Terraform for a greenfield data lake-house platform, expanding my skill-set into AWS and cloud technologies.

Consulting & Personal Projects

Consulted on cloud architecture for a AML transaction monitoring platform.

Built open-source components for managing VYoS routers at scale.

[terraform-provider-vyos](#) 

University of Liverpool

BSc (Hons) Computer Science (First Class)